

1. Administrative & Study Identification

Full Study Title: A PROSPECTIVE, RANDOMIZED, OPEN-LABEL PHASE 2 STUDY TO EVALUATE THE SUPERIORITY OF INOTUZUMAB OZOGAMICIN MONOTHERAPY VERSUS ALLR3 FOR INDUCTION TREATMENT OF CHILDHOOD HIGH RISK FIRST RELAPSE B-CELL PRECURSOR ACUTE LYMPHOBLASTIC LEUKAEMIA **Short Title/Acronym:** InO vs ALLR3 in Pediatric HR First Relapse ALL **Protocol Number:** B1931036 **NCT Number:** NCT05748171 **Posting ID:** 987429954398390850 **Sponsor/Company:** Pfizer **Investigational Product:** Inotuzumab ozogamicin (InO) / Preferred UMLS Name: Inotuzumab Ozogamicin **Trade Name:** Besponsa **ATC Code:** L01FB01 **EMA Product Information:** [EMA URL](#) **Phase of Development:** Phase 2 **Trial Status:** RECRUITING (Currently recruiting participants) **Medical Monitor:** Pfizer Clinical Operations / CT.gov Call Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the superiority of Inotuzumab ozogamicin (InO) monotherapy versus the standard ALLR3 (UKALL-R3) chemotherapy regimen after 1 cycle of induction treatment in pediatric participants with High Risk (HR) first bone marrow relapse of CD22-positive B-cell precursor acute lymphoblastic leukemia (BCP ALL). **Secondary Objectives:** To evaluate the safety, tolerability, pharmacokinetics (PK), and long-term efficacy (including Event Free Survival (EFS), Duration of Response (DoR), and Rate of hematopoietic stem cell transplantation (HSCT)) of InO in this patient population.

2.2 Background & Rationale Children with high-risk, first-relapse BCP ALL face poor outcomes with standard intensive chemotherapy regimens, which are often associated with high toxicity. Inotuzumab ozogamicin, a CD22-directed antibody-drug conjugate (ADC), has demonstrated high efficacy in achieving complete remission and minimal residual disease (MRD) negativity in relapsed/refractory ALL. This study aims to determine if utilizing InO monotherapy during the induction phase can provide superior efficacy and a safer bridge to consolidation or hematopoietic stem cell transplantation (HSCT) compared to the standard ALLR3 chemotherapy block.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (ALLR3 Chemotherapy) **Blinding:** Open-label **Randomization:** 2:1 ratio (InO Monotherapy : ALLR3 Chemotherapy)

3.2 Planned Enrollment & Duration Target Enrollment (\$N\$): 100 participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 17-May-2023 **Estimated Primary Completion:** 04-Nov-2036 **Follow-up Duration:** Treatment will end after induction therapy; follow-up for efficacy and safety will continue for up to 5 years from randomization.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants between 1 and <18 years of age. 2. Morphologically confirmed diagnosis of first relapse HR BCP ALL; HR first relapse is defined as relapse occurring within 18 to 30 months of original diagnosis of ALL or within 6 months of completion of primary therapy. 3. CD22-positive ALL (as defined by local institution) with bone marrow involvement of $\geq 5\%$ leukemic blasts ($\geq$ M2 status). 4. Adequate organ function: eGFR (in participants 1 to <2 years) or eCrCl (in participants 2 to <18 years) ≥ 30 mL/min; AST and ALT $\leq 5 \times$ ULN; Total bilirubin $\leq 1.5 \times$ ULN (unless documented Gilbert's syndrome). 5. Prior history of thrombosis during corticosteroid and/or asparaginase use is permitted provided the patient receives anti-coagulant prophylaxis per institutional guidelines. 6. Cardiac shortening fraction $\geq 30\%$ by echocardiogram or ejection fraction $>50\%$ by MUGA. 7. Participants with combined bone marrow and testicular relapse are eligible assuming orchiectomy is performed prior to randomization or is planned at the end of induction therapy.

4.2 Exclusion Criteria 1. Presence of identified very high-risk genetic abnormalities (e.g., KMT2A::AFF1, TCF3-HLF, TCF3-PBX1, hypodiploidy < 40 chromosomes or masked low hypodiploidy, TP53 alteration) or Philadelphia-chromosome positive ALL (BCR-ABL/t(9;22) present). 2. Any history of prior or ongoing hepatic veno-occlusive disease (VOD) / sinusoidal obstruction syndrome (SOS), or prior liver failure [defined as severe acute liver injury with encephalopathy and impaired synthetic function (INR of ≥ 1.5)]. 3. Prior allogeneic HSCT or CAR T-cell therapy. 4. Prior therapy with a calicheamicin-conjugated antibody (e.g., InO or gemtuzumab ozogamicin). 5. Isolated extramedullary leukemia. 6. Participants with active, uncontrolled bacterial, fungal, or viral infection. 7. Hypersensitivity/allergy to both PEG-ASP and Erwinia-ASP.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Inotuzumab ozogamicin (InO) monotherapy administered intravenously, or standard ALLR3 (dexamethasone, vincristine, mitoxantrone, PEG-asparaginase) block 1 induction chemotherapy. **Route of Administration:** Intravenous (IV) **Duration of Treatment:** 1 cycle

of induction therapy. End of Treatment is defined as occurring upon recovery from 1 cycle of study therapy (Day 28 $\pm$ 2 days), or one day before initiation of new anticancer therapy, whichever occurs first. **Post-Induction:** Responding participants are expected to proceed to standard of care (SOC) consolidation therapy upon recovery of blood counts, but no sooner than 7 days after the last dose of study intervention. Non-responders are expected to proceed with salvage therapy at the investigator's discretion.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care; anticoagulant prophylaxis for those with a prior history of thrombosis. **Prohibited:** Concurrent investigational agents, targeted therapies for very high-risk genetic abnormalities, or live viral vaccines during the induction phase.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Bone Marrow Aspirate (CD22 and blast % check), Organ Function Labs, Cardiac Echo/MUGA
Baseline	Day 0	Randomization (2:1), First Dose of Study Intervention
Interim	Cycle 1 (Days 1-28)	Safety Labs, PK Sampling, Assessment of DLTs / Hepatic panels
End of Induction	Day 28 $\pm$ 2	Bone Marrow Assessment for CR/CRp/CRi and MRD negativity
End of Study	Up to 5 Years	Long-term follow-up for all participants (responders and non-responders) for EFS, OS, DoR, and HSCT outcomes

7. Outcome Measures (Endpoints)

7.1 Primary Outcome Definition: Minimum Residual Disease (MRD) Negativity in participants achieving complete response (CR), complete response with incomplete platelet count recovery (CRp), or complete response with incomplete count recovery (CRi).

Measurement Method: MRD negativity status is determined based on the minimum MRD percentage between the date of CR/CRp/CRi and end of treatment test as assessed by RQ-PCR, with reflex to Flow Cytometry (FC) result if MRD is non-evaluable by RQ-PCR.

7.2 Secondary Outcomes

- * **Event Free Survival (EFS):** Time from randomization to the first occurrence of an event.
- * **Duration of Response (DoR):** Evaluated for participants who achieved CR/CRp/CRi.
- * **Rate of hematopoietic stem cell transplantation (HSCT):** Proportion of participants proceeding to HSCT.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits and laboratory assessments, specifically tracking liver function tests (AST/ALT/Bilirubin) to monitor for sinusoidal obstruction syndrome (SOS) / veno-occlusive disease (VOD). **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (IDMC) to review unblinded safety and efficacy data, particularly given the historical risk of hepatotoxicity with ADCs.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all participants receiving at least one dose of the assigned treatment. **Sample Size Justification:** Target N=100, randomized 2:1 (InO to ALLR3), powered to detect a statistically significant improvement in the rate of MRD-negative CR following 1 cycle of induction therapy. **Handling of Missing Data:** Participants withdrawing or missing the End of Induction assessment will be treated as non-responders for the primary endpoint. Survival data will be right-censored at the date of last contact.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol adherence, safety reporting, and accurate documentation of bone marrow blast percentages and MRD levels. **Audit Trail:** Robust electronic Case Report Form (eCRF) tracking with data clarification processes and centralized review of critical laboratory results.

11. Study Identifiers & Governance

Primary ID: B1931036 **Registry Number:** NCT05748171 **Posting ID:** 987429954398390850 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study to Learn More About the Study Medicine Called Inotuzumab Ozogamicin (InO) in Children (1 to <18 Years) With First Relapse ALL **Official Regulatory Title:** A PROSPECTIVE, RANDOMIZED, OPEN-LABEL PHASE 2 STUDY TO EVALUATE THE SUPERIORITY OF INOTUZUMAB OZOGAMICIN MONOTHERAPY VERSUS ALLR3 FOR INDUCTION TREATMENT OF CHILDHOOD HIGH RISK FIRST RELAPSE B-CELL PRECURSOR ACUTE LYMPHOBLASTIC LEUKAEMIA

12. Clinical Framework (Oncology / Leukemia Specific)

Disease Area: ACUTE LYMPHOBLASTIC LEUKEMIA **Preferred Name:** Acute lymphocytic leukemia **Diagnosis Threshold:** First bone marrow relapse HR BCP ALL with $\geq 5\%$ leukemic blasts and CD22-positivity **Medical Methodology:** Targeted Antibody-Drug Conjugate (ADC) versus intensive combination chemotherapy **MeSH Code:** D007936 **SNOMED ID:** 128822004 **Optimization Target:** CR/CRp/CRi with Minimum Residual Disease (MRD) negativity

13. Study Procedures & Methodology

Management Pathway: Induction therapy followed by standard of care consolidation or HSCT **Education Protocol (HCP):** Mandatory training on VOD/SOS mitigation and monitoring, and management of cytopenias **Education Protocol (Patient):** Informed consent/assent outlining the mechanism of targeted ADCs vs. standard chemotherapy **Quality Audit Mechanism:** Centralized laboratory verification of CD22 expression and MRD assessment (via RQ-PCR and FC reflex)

14. Observation & Follow-Up Schedule

Total Duration: Up to 5 years from randomization **Onsite Visit Cadence:** Frequent monitoring during the 28-day induction cycle; subsequent visits per standard oncology follow-up **Remote Monitoring Frequency:** Routine survival tracking post-induction phase **Checklist Review Interval:** End of Induction (Day 28 $\pm$ 2) response assessment

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				